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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

RI2225-1 – NATURAL LANGUAGE PROCESSING
(For RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the main challenge posed by lexical ambiguity in NLP?**
A) Different spellings for the same word B) The same word having multiple possible meanings
C) Difficulty in tokenization D) Unknown words in a text
2. **What do Conditional Random Fields (CRFs) model?**
A) Joint probability distribution over hidden states B) Conditional probability distribution of output given input
C) Probability of observing data D) The likelihood of training data
3. **What is the Baum-Welch algorithm primarily used for in the context of HMMs?**
A) Decoding hidden states B) Estimating the parameters of an HMM when the training data is unlabeled
C) Tokenization D) POS tagging
4. **Which of these is considered a foundational step in text processing?**
A) Image segmentation B) Tokenization
C) Data encryption D) Signal modulation
5. **Which of these techniques is used to reduce words to their root form by removing suffixes and prefixes?**
A) Lemmatization B) Stemming
C) Tokenization D) Parsing
6. **What specific problem does the Viterbi algorithm solve in the context of Hidden Markov Models (HMMs)?**
A) Parameter estimation B) Finding the most probable sequence of hidden states
C) Smoothing probabilities D) Feature selection
7. **What is the core objective of Natural Language Processing (NLP)?**
A) Processing numerical data B) Processing human language to enable computers to understand and interact with it
C) Image recognition D) Audio signal processing
8. **What is the primary objective of Part-of-Speech (POS) tagging?**
A) To identify the main topic of a document B) To assign a grammatical category tag to each word in a text
C) To correct spelling errors D) To generate summaries
9. **How does WordNet contribute to understanding lexical relationships?**
A) By parsing sentences B) By defining word relations in a lexical database
C) By translating languages D) By summarizing texts

10. **What is a key aspect of Transition-Based Parsing?**
 A) Converting sentences to Chomsky Normal Form
 B) Using labeled data to learn transitions between configurations
 C) Calculating inside-outside probabilities
 D) Building phrase structure trees
11. **What is the main aim of Transition-Based Parsing?**
 A) Converting sentences to CNF
 B) Finding the transitions to find the dependency path for a new sentence
 C) Calculating inside-outside probabilities
 D) Building phrase structure trees
12. **What is the purpose of parsing in NLP?**
 A) To generate text
 B) To analyze the syntactic structure of sentences
 C) To understand the meaning of words
 D) To correct spelling errors
13. **How do word embeddings capture semantic meaning?**
 A) By listing synonyms
 B) By converting words into numerical vectors
 C) By defining grammatical rules
 D) By translating languages
14. **In MST-Based Dependency Parsing, what does MST stand for?**
 A) Maximum Spanning Tree
 B) Minimum Syntactic Translation
 C) Most Significant Term
 D) Minimal Semantic Transfer
15. **What role does WordNet play in Lexical Semantics?**
 A) It is a parsing algorithm
 B) It is a lexical database that defines relations between words
 C) It is a machine translation tool
 D) It is a sentiment analysis lexicon
16. **What is the focus of Novel Word Sense Detection?**
 A) Identifying the origin of words
 B) Discovering new meanings of existing words
 C) Correcting spelling errors
 D) Translating text into different languages
17. **What is Information Extraction (IE) in NLP?**
 A) Extracting structured information from unstructured text
 B) Translating text into different languages
 C) Summarizing text
 D) Correcting grammatical errors
18. **What is the focus of Relation Extraction?**
 A) Extracting only the names of entities
 B) Identifying relationships between entities in text
 C) Summarizing text
 D) Correcting grammatical errors
19. **What is the main goal of Topic Models in NLP?**
 A) To translate text into different languages
 B) To discover the underlying topics in a collection of documents
 C) To correct grammatical errors
 D) To summarize text
20. **What is the primary goal of Text Classification?**
 A) To translate text
 B) To assign predefined categories to text documents
 C) To summarize text
 D) To correct grammatical errors

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define edit distance with an example.	4	L*2	1	1
	b) Differentiate between unigrams and bigrams with examples.	4	L2	1	1
	c) How is the Maximum Entropy Model different from HMM?	8	L4	2	1
2.	a) Explain the role of smoothing in language models? Briefly explain Laplace smoothing.	8	L4	2	1
	b) Define Stemming and Lemmatization. What are the differences between Lemmatization and Stemming?	8	L2	1	1
3.	a) Discuss the differences between supervised and unsupervised POS tagging methods.	8	L4	2	1
	b) Compare and contrast Laplace and Good-Turing smoothing methods.	8	L2	1	1

Unit – II

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|-------|---|---|----|---|---|
| 4. a) | What is Probabilistic Context-Free Grammars (PCFGs) and how are they used in parsing? | 8 | L2 | 3 | 1 |
| b) | What is lexical semantics? Discuss its importance in NLP. | 8 | L2 | 4 | 1 |
| 5. a) | Explain dependency grammars and how they are used in parsing natural language. | 8 | L2 | 3 | 1 |
| b) | Discuss the advantages and limitations of word embeddings in NLP. | 8 | L4 | 4 | 1 |
| 6. a) | Explain the difference between supervised and unsupervised parsing techniques. | 8 | L4 | 3 | 1 |
| b) | Compare and contrast the Skip-gram and CBOW models used in word embeddings. | 8 | L4 | 4 | 1 |

Unit – III

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|-------|---|---|----|---|---|
| 7. a) | What are the different LDA variants and their applications. | 8 | L4 | 5 | 1 |
| b) | Explain the evaluation techniques used for text summarization. | 8 | L3 | 5 | 1 |
| 8. a) | How does entity linking enhance the performance of information retrieval systems? | 8 | L2 | 5 | 1 |
| b) | Explain the differences between rule-based and machine learning-based information extraction methods. | 8 | L4 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
